PERSONAL EXPENSE TRACKER

Python Essentials — Project Report

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Course: Python Essentials

Project Type: Command-Line Application

1. Introduction

Managing daily expenses manually can make it difficult to understand where money is being spent. The Personal Expense Tracker is a command-line Python application designed to record and analyze personal expenses. The application stores expense information locally in JSON format and provides options for searching, deleting, summarizing, and exporting records.

2. Problem Statement

People often record expenses in notebooks or scattered applications, making it difficult to search previous transactions or calculate monthly spending. This project provides a lightweight terminal-based solution for recording, searching, deleting, analyzing, and exporting expenses.

3. Objectives

Develop a practical Python command-line application; demonstrate core Python concepts; implement persistent file storage; validate user input; generate useful statistics; and provide CSV export functionality.

4. Scope

The application includes expense creation, listing, searching, deletion, monthly analysis, category-wise analysis, CSV report generation, and JSON persistence. Online banking, cloud synchronization, authentication, and payment processing are outside the scope.

5. Technologies Used

Python 3, JSON, CSV, datetime, pathlib, uuid, and collections. The project uses only Python standard-library modules and requires no third-party packages.

6. System Design

Program flow: Start → Load expenses.json → Display menu → Read user choice → Perform selected operation → Save/display results → Return to menu or exit.

7. Module Description

main.py manages the CLI menu and user interaction. expense_manager.py contains the ExpenseManager class for storage, CRUD operations, summaries, and CSV export. utils.py contains reusable validation and display functions. expenses.json provides persistent local storage.

8. Functionalities

The system can add expenses, view records, search by category or description, delete records by ID, calculate monthly totals/averages/highest spending, display category-wise spending, and export records to CSV.

9. Input Validation

The application validates numeric amounts, rejects zero or negative amounts, prevents blank category/description values, validates DD-MM-YYYY dates, and handles invalid menu choices.

10. Testing

Testing covers adding valid expenses, invalid amounts, negative amounts, blank input, viewing records, searching, deleting valid and invalid IDs, monthly summaries, and CSV export. The included TESTING.md contains step-by-step test cases.

11. Advantages

Simple terminal interface; no external database; no third-party dependencies; persistent storage; input validation; modular code; CSV export; easy execution.

12. Limitations

Data is stored locally on one computer, there is no authentication or GUI, there is no cloud backup, and the application is intended for personal rather than multi-user use.

13. Future Enhancements

Possible enhancements include a GUI, SQLite database, authentication, budget alerts, charts, recurring expenses, and cloud synchronization.

14. Conclusion

The Personal Expense Tracker demonstrates how Python can be used to build a practical command-line application. It combines functions, classes, collections, file handling, JSON, CSV, validation, exception handling, and date processing in one executable project.